

PROJECT TASK: GST BILLING SYSTEM

Mobile / Web Application Development Practical Task

1. Objective

Design and develop a GST Billing System that allows a shopkeeper to manage customers (parties), create GST-compliant invoices for goods or services sold, automatically calculate applicable taxes, and generate a downloadable/printable PDF invoice. The project should reflect real-world billing workflows used by small and medium retail businesses.

2. Problem Statement

Small shopkeepers currently prepare bills manually or using spreadsheets, which is time-consuming and error-prone, especially when calculating GST (CGST, SGST, IGST) across multiple items. Build an application that lets a shopkeeper maintain a party (customer) list, create itemized GST bills against a selected party, automatically compute tax and totals, save bill history, and export/share each bill as a PDF.

3. Core Modules & Functional Requirements

3.1 Party (Customer) Management

- Add a new party with details: Name, Mobile Number, Address, State, GSTIN (optional), Email (optional)
- Edit and delete an existing party
- View a searchable list of all parties
- View a single party's bill history

3.2 Item / Product Management

- Add items/products with: Item Name, HSN/SAC Code (optional), Unit Price, GST %
- Edit and delete items
- Maintain a reusable item list so items don't need to be retyped for every bill

3.3 GST Bill Creation

- Select a party for the bill (or quick-add a new one)
- Add one or more items to the bill with quantity
- Auto-calculate, per item: Taxable Amount = Rate $\times$ Quantity
- Auto-calculate GST split:
 - Same state $\rightarrow$ CGST + SGST (split the item's GST % equally)
 - Different state $\rightarrow$ IGST (full GST % as one tax)
- Auto-calculate: Subtotal, Total Tax, Grand Total (rounded appropriately)
- Auto-generate a unique, sequential Invoice Number and Invoice Date
- Save the bill; saved bills must not be editable after generation (create a credit note / new bill for corrections)

3.4 PDF Invoice Generation

- Generate a professional PDF invoice containing:
 - Business/shop header (name, address, GSTIN)
 - Invoice number, date, and party (bill-to) details

- Itemized table: item, quantity, rate, taxable amount, GST %, CGST/SGST or IGST, line total
- Subtotal, total tax, and grand total (in figures; words is a bonus)
- Allow the PDF to be shared (WhatsApp/email) or saved to device/downloaded

3.5 Bill History & Dashboard

- List of all bills with date, party name, and amount, sorted by most recent
- Search/filter bills by party name or date range
- Simple dashboard: total sales, total tax collected, number of bills (today / this month)

4. Suggested Data Model

Entity	Key Fields	Notes
Party	id, name, mobile, address, state, gstin, etc	One shopkeeper can have many parties
Item	id, name, hsnCode, price, gstPercent, etc	Reusable across bills
Bill	id, invoiceNo, date, partyId, items[], subtotal, totalTax, grandTotal, etc	items[] is a list of BillItem
BillItem	itemId, name, qty, rate, gstPercent, taxableAmt, cgst, sgst, igst, etc	Snapshot values at time of billing

5. GST Calculation Logic (Reference)

1. Taxable Amount = Rate $\times$ Quantity
2. If Party State = Shop's State $\rightarrow$ CGST = SGST = (GST% $\div$ 2) of Taxable Amount
3. If Party State $\neq$ Shop's State $\rightarrow$ IGST = GST% of Taxable Amount
4. Line Total = Taxable Amount + Tax Applied
5. Grand Total = Sum of all Line Totals

6. Suggested

- PDF generation: pdf / printing packages (Flutter) or a PDF library of choice (Web)
- Optional: Firebase/Supabase for cloud sync of parties and bills

7. Deliverables

1. Working application (source code, buildable/runnable)
2. At least 3 sample bills generated as PDF, included in the submission
3. Short report (2–3 pages): modules covered, tech stack used, screenshots

8. Bonus / Stretch Features (Optional)

- Discount field per item or per bill
- Payment status tracking (Paid / Unpaid / Partial)
- Multiple GST rate slabs picker (0%, 5%, 12%, 18%, 28%)
- Export bill list as Excel/CSV in addition to per-bill PDF
- Barcode/QR-based item entry
- Multi-user login for the shop (staff accounts)

9. Submission Guidelines

- Submit source code as a GitHub repository link or a zipped project folder
- Include the report and sample PDF invoices in the same submission
- Ensure the project runs from a clean clone/setup before submission

— *End of Task Document* —